ABHRONEEL MUKHERJEE

Applied AI & Machine Learning Engineer | Quantitative Intelligence & Autonomous Robotics

Pune, Maharashtra, India | abhro00@gmail.com | +91-9831867527 | github.com/quantroninblue | LinkedIn: Abhroneel Mukherjee

PROFESSIONAL SUMMARY

Applied AI and machine learning engineer building end-to-end intelligent systems across quantitative finance and autonomous robotics. Architected Vol Engine, a quantitative decision system for SPX options research, and Odysseus, an adaptive navigation intelligence stack with causal failure reasoning and recovery. Also created Hephaestus, an open-source AI robotics workbench connecting natural-language orchestration with CAD, Gazebo, and ROS 2.

TECHNICAL SKILLS

Programming: Python, C++, SQL, Bash, R
Machine Learning: PyTorch, scikit-learn, XGBoost, GMMs, HMMs, multi-task learning, time-series validation, causal and probabilistic modeling, model evaluation, model governance
Applied AI & NLP: LLM APIs, NLP, prompt engineering, structured outputs, function/tool calling, agent orchestration, Retrieval-Augmented Generation (RAG), embeddings, vector search, LangChain, LangGraph, Hugging Face Transformers
Robotics & Autonomous Systems: ROS 2, Gazebo Harmonic, CadQuery, OpenCV, SLAM/VSLAM, RGB-D perception, sensor fusion, Kalman filtering, probabilistic localization, occupancy mapping, A*, path planning and control
ML Systems & Backend: FastAPI, Pydantic, Docker, Kubernetes, Linux, Git, GitHub Actions/CI, AWS, REST APIs, PostgreSQL, relational databases, vector databases, dataset versioning, experiment tracking, model serving, CUDA, ONNX
Scientific Computing: NumPy, Pandas, SciPy, Statsmodels

PROFESSIONAL EXPERIENCE

Machine Learning Researcher | Vol Engine — Jun 2026 – Present

RiskOptima WealthTech | *Remote · Part-time alongside MSc*

- Architecting Vol Engine, a production-oriented quantitative ML research system for systematic SPX options decisions, spanning market-data ingestion, feature engineering, volatility forecasting, regime detection, execution-cost modeling, risk, and portfolio construction.
- Built the forecasting stack around realized and implied volatility, skew, term structure, and regime features; used GMM/HMM state models and XGBoost realized-volatility forecasts to inform candidate selection.
- Designed a separate multi-task neural meta-label layer estimating win probability, expected return, and downside risk, with confidence-aware abstention and mandatory human review before any capital moves.
- Encoded transaction costs, margin, buying power, Greeks, and portfolio constraints into a narrow PRESCRIPTION/NO_TRADE decision contract suited to 0-DTE research workflows.
- Implemented versioned datasets and artifacts, resumable checkpoints, experiment tracking, purged walk-forward validation, out-of-distribution safeguards, replayable dry-run research, and 182 automated tests.

Computer Vision & Robotics Intern | Odysseus — Summer 2026

Vector Robotics | *Pune · Summer internship during MSc*

- Architected Odysseus, an adaptive embodied navigation intelligence architecture integrating perception, probabilistic localization, planning, adaptive recovery, and persistent spatial memory on a ROS 2 autonomy stack in Python and C++, validated in Gazebo.
- Developed a bounded local trajectory planner evaluating 63 candidate rollouts per control cycle against collision risk, clearance, heading alignment, and goal progress, with a deterministic runtime safety override.
- Built a PyTorch causal reasoning model that classifies outcomes across 11 failure and success causes and predicts recovery confidence, enabling targeted recovery rather than a single fallback behavior.
- Engineered persistent spatial memory for hazardous regions and reliable navigation corridors, plus retracing and exploration policies that adapt behavior across deployments.
- Integrated semantic segmentation, depth-aware estimation, Kalman multi-object tracking, probabilistic localization, local and persistent occupancy maps, and automated validation covering perception, planning, and safety.

OPEN-SOURCE ENGINEERING

Creator & Maintainer | Hephaestus AI — Jul 2026 – Present

Open Source | *Python · CadQuery · ROS 2 Jazzy · Gazebo Harmonic*

GitHub: github.com/quantroninblue/Hephaestus | PyPI: pypi.org/project/hephaestus-ai

- Created and released an AI-native robotics workbench that turns natural-language design intent into editable CadQuery source, STEP/STL artifacts, Gazebo simulations, and ROS 2 workflows.
- Designed a typed tool-orchestration runtime with workspace-constrained atomic edits, human approval gates, deterministic validation, bounded subprocess execution, and session-owned process cleanup.
- Built and verified a differential-drive vertical slice spanning URDF/Xacro, SDF, ros_gz_bridge, gz_ros2_control, LiDAR, IMU, joint states, odometry, controller activation, and velocity-command motion in simulation.
- Published hephaestus-ai 0.1.1 to PyPI with pipx installation, 361 automated tests, protected GitHub Actions CI, reproducible wheel/sdist builds, and Trusted Publishing.

EDUCATION

MSc Economics | Data Analytics Specialisation — 2025 – 2027

Symbiosis School of Economics

BA (Honours) — 2020 – 2024

Symbiosis School of Liberal Arts